A Mechanistic, ATP-Dependent Cross-Bridge Model of Cardiac Muscle Mechanics Under Energetic Stress

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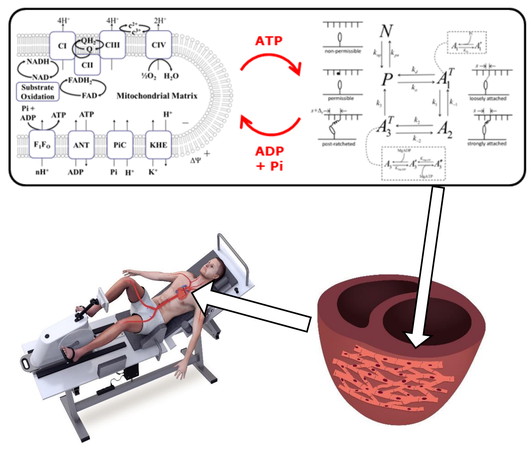
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INTRODUCTION

- Heart failure is accompanied by an energetic deficit: impaired mitochondrial function and reduced adenine nucleotide pools lead to lower [ATP] higher [Pi], and reduced PCr/ATP ratio [1]
- These changes directly inhibit the myosin cross-bridge cycling — the molecular engine of cardiac contraction
- Beard et al. 2022 [2] established a strain-discretized XB model fit to FV data — but does not reproduce ATP-dependent contractile differences
- Closing this gap requires richer experimental data and additional physical mechanisms in the model



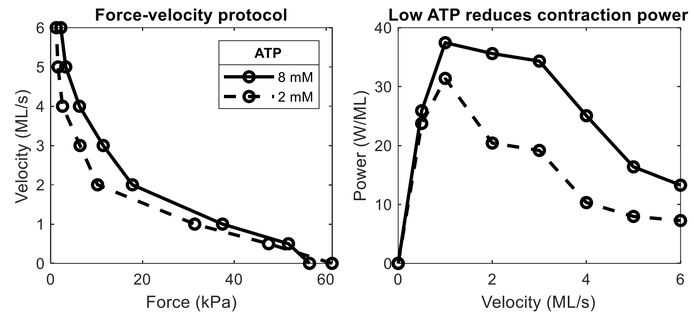
MAIN QUESTIONS

- How does [ATP] modulate sarcomere-level force, velocity, and power stroke kinetics?
- Which specific cross-bridge mechanisms can account for the ATP sensitivity?
- Can a single mechanistic model simultaneously reproduce force-velocity and slack-restretch behavior across ATP concentrations?

ATP MODULATES MUSCLE POWER

Replication of Beard et al. 2022

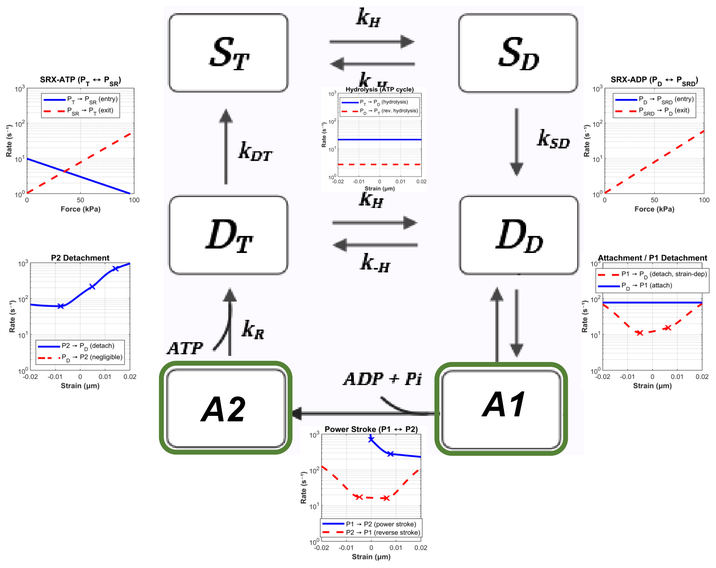
- Reduced [ATP] slows XB detachment → more attached heads → higher isometric force but slower cycling and reduced peak power output
- FV curve shifts with [ATP]: lower ATP → higher F_0, lower Vmax, reduced peak power
- Beard 2022 model reproduces individual FV curves but does not capture ATP-dependent differences in contractile phenotype
- A nucleotide-explicit state representation is needed to capture [ATP]- and [ADP]-dependent rate changes mechanistically



MODEL

Six-state strain-discretized probabilistic cross-bridge (XB) framework

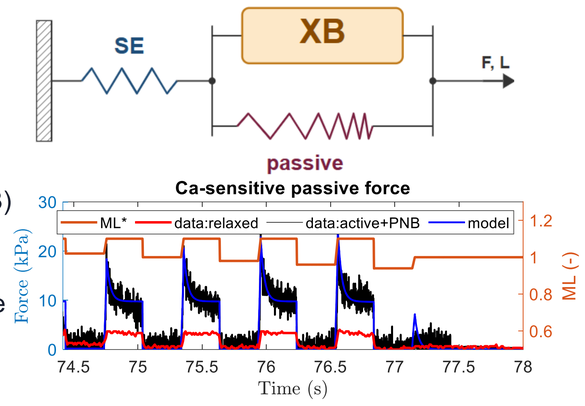
- [ATP], [ADP], and [Pi] modulate transition rates biochemically throughout the full cycle
- Asymmetric piecewise strain-dependent modifiers on attached-state transition rates
- Force-dependent SRX mobilization encodes thick filament mechanosensing (Frank-Starling)
- ATP-dependent effect hypothesis:
  - A2 detachment, k_H and k_{-H}



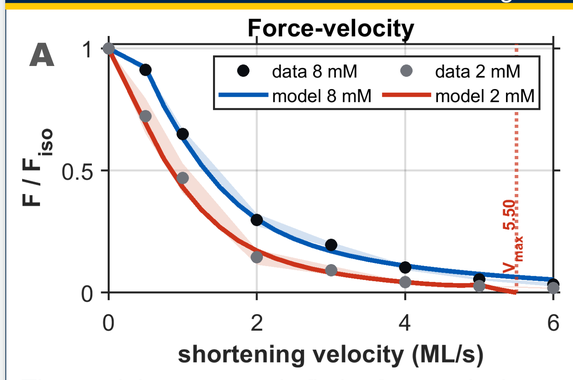
Pool	States	Role
Disordered Relaxed (DRX)	**D_T, D_D**	Cycling-ready free heads, ATP-bound and ADP·Pi-bound
Attached	**A1 (pre), A2 (post-stroke)**	Strain-sensitive, force-generating
Super-Relaxed (SRX)	**S_T, S_D**	Inhibited folded heads, ATP-bound and ADP·Pi-bound

Mechanical configuration and Ca-sensitive passive component

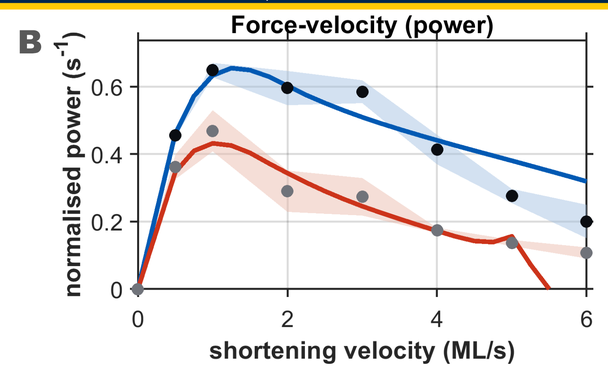
- The crossbridge (XB) element is coupled with parallel passive and a general serial element
- In relaxed condition (no Ca^{2+}) the stress peaks at ~3 kPa following stretch
- In high-Ca^{2+} condition (with PNB+Mava deactivating the XB) the passive force peaks at ~20 kPa — a phenomenon attributed to Ca^{2+}-mediated effects on titin [3]
- A single-exponential decay captures the amplitude and time course sufficiently for this purpose; for a detailed passive model see [3].



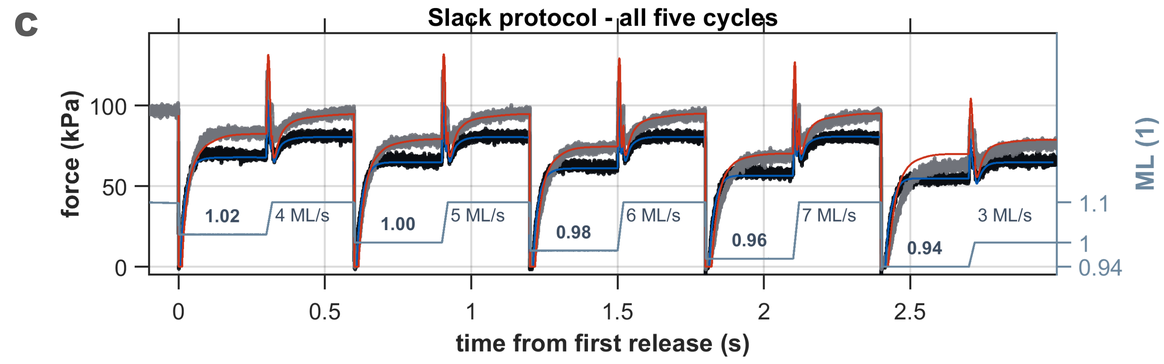
RESULTS – all data from mouse right-ventricular trabeculae, 8 mM vs 2 mM ATP



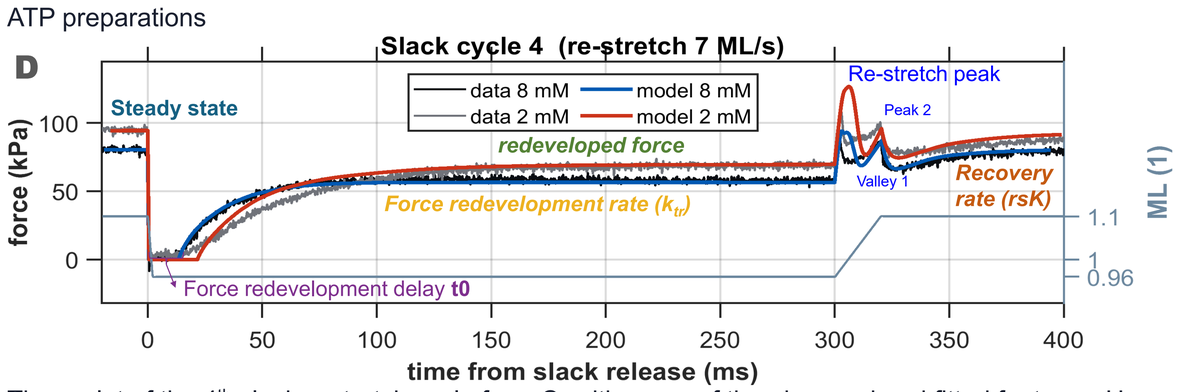
The model can correctly fit the force velocity profile for both ATP levels, with minor instability at higher speeds



The replot of the same data as A reveals larger differences in ATP levels. Yet, the model is still able to satisfactorily reproduce the power-velocity profile.

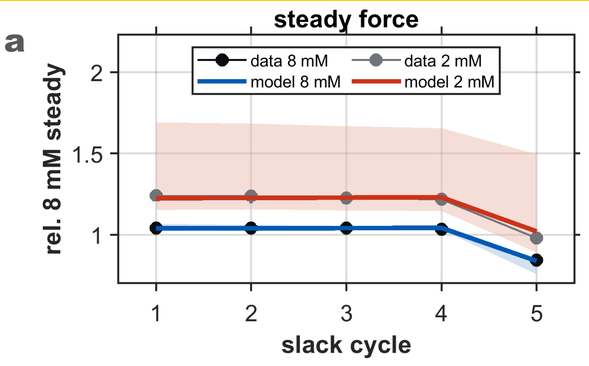


For the slack-restretch protocol, the muscle length (ML, normalized to SL of 2.0 μm) was rapidly slackened, held and restretched by various velocities five-times. Model matches both high- and low-ATP preparations

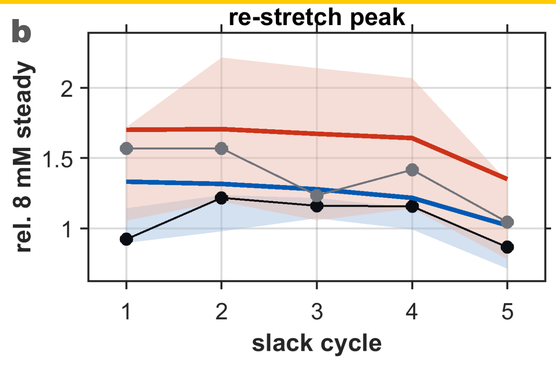


The replot of the 4th slack-restretch cycle from **C**, with some of the observed and fitted features. Here, some of the discrepancy is visible, yet the model can still distinguish the high- and low- ATP well.

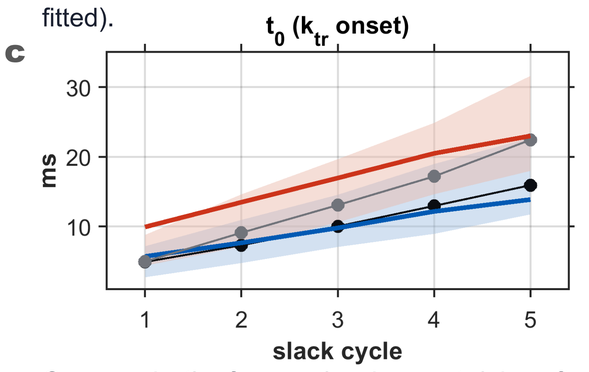
SLACK FEATURES from panel C over all 5 slack-restretch cycles



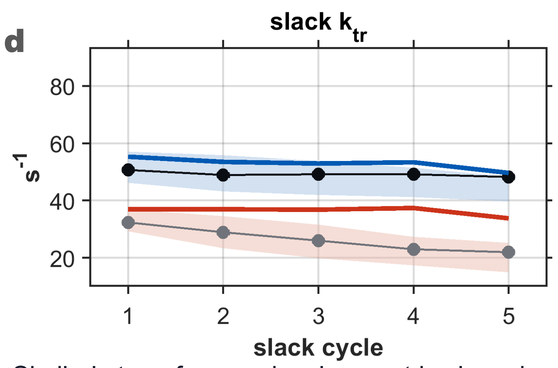
The model perfectly reproduces the relative ATP effect on steady-state force of the identified dataset. Shaded band = SD of 3 preparations (not fitted).



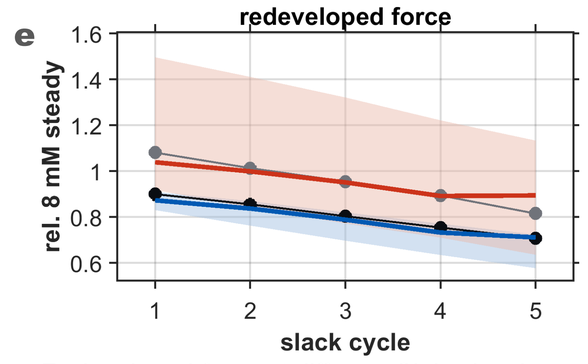
First re-stretch peak feature shows slight overestimation by the model, consistently for high- and low ATP.



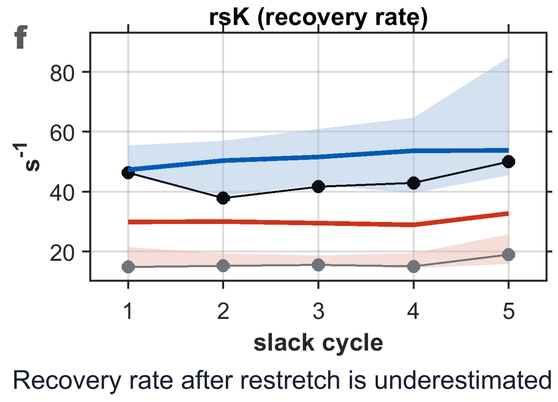
Conversely, the force redevelopment delay after slack is increased in low-ATP. The model still reproduces the slopes for different slack lengths.



Similarly to ***c***, force redevelopment is slowed down for low ATP. The model tracks the decrease, but to limited extent.



Redeveloped force at different SL is closely tracked for both ATP levels.



Recovery rate after restretch is underestimated for low ATP levels, yet the direction is correct.

CONCLUSION

- Low ATP consistently increases force-related features (steady state, restretch peaks) by about 20% and slows down the contraction (t0 delay +30%, ktr -45%, normalized power -28%).
- A six-state XB model closely reproduces all key features of muscle dynamics for both high- and low- ATP levels
- Framework provides a mechanistic bridge between sarcomere mechanics and cardiac energetics

FUTURE WORK

- Additional measurements to constrain the spread estimate.
- Further quantitative parameter optimization for low-ATP
- Testing of other potentially sensitive ATP, ADP and Pi gates
- Explicit coupling to mitochondrial ATP production model

LOW ATP MAKES MUSCLE CONTRACT STRONGER, BUT SLOWER. THE MODEL REPRODUCES BOTH.

References

[1] Collins NL, Dasika S, Van den Bergh F, Bazil J, Beard DA. Systems Analysis of Carboxylate Transport and Oxidation Pathways in Cardiac Mitochondria. bioRxiv 2026.
[2] Beard DA et al. Reduced cardiac muscle power with low ATP simulating heart failure. Biophys. J. 2022.
[3] Jezek F et al. Theoretical analysis of power-law stress relaxation and calcium-dependent passive mechanics in cardiac muscle. J. Physiol. 2026.

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